

Introduction to Object Oriented Programming (OOP)

CSC 211

Course Duration: 10 Weeks

Course Objectives:

- Introduction to Object Oriented Programming (OOP)
- Overview of Java as a programming language.

Java Environment Setup

- **Java Environment Setup**

- Installing JDK, JRE, and IDE (e.g., IntelliJ, Eclipse).
- Introduction to Java Development Kit (JDK), Java Runtime Environment (JRE), and Integrated Development Environment (IDE).

First Java Program

First Java Program

- Writing and running a simple Java program (Jeff, Geoff !!!).
- Understanding the structure of a Java program.

Week 2: Basic Java Syntax and Data Types

•Java Syntax

- Variables and data types (int, double, char, String, boolean).
- Type casting and type conversion.

Operators

- **Operators**

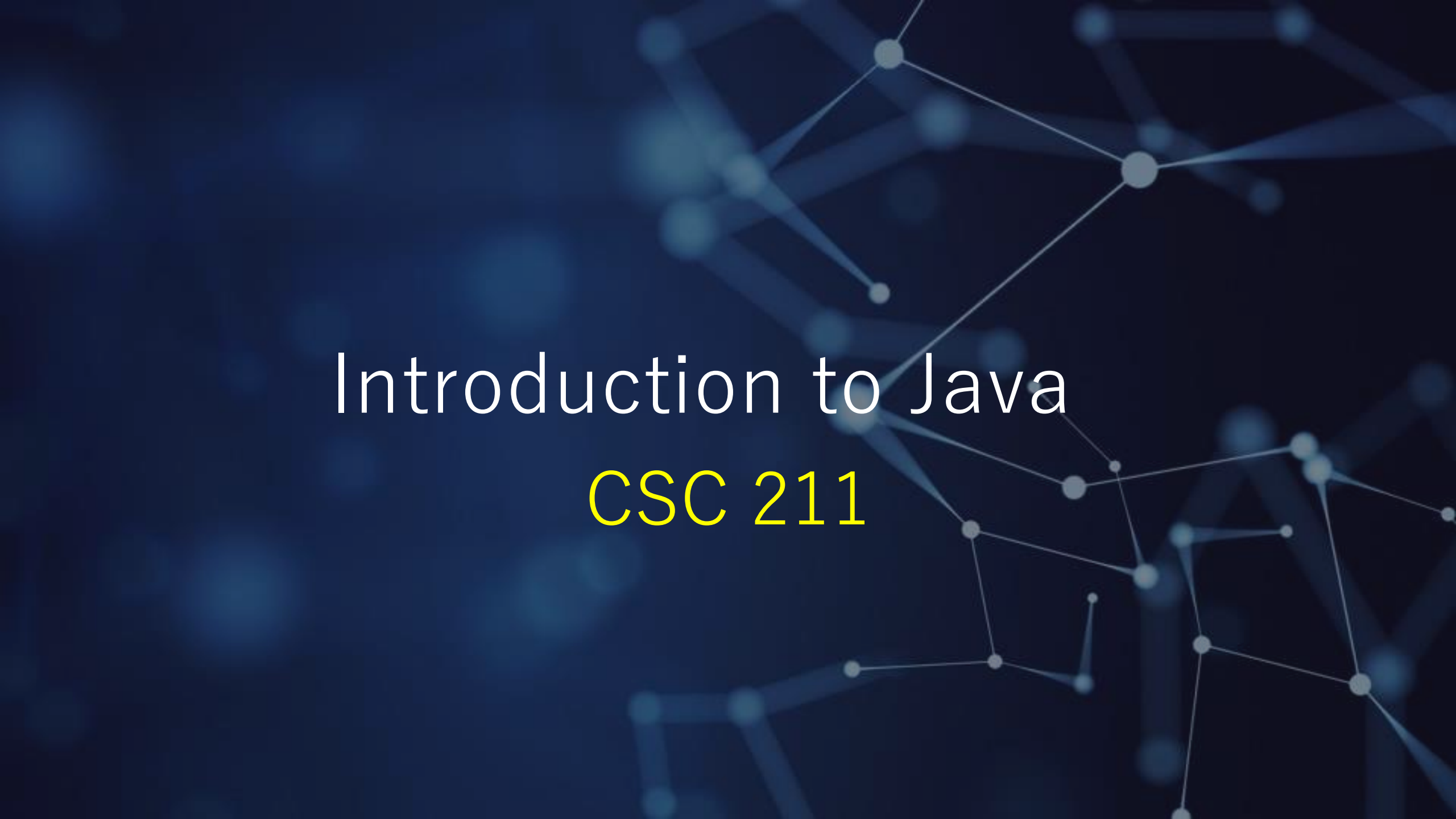
- Arithmetic, relational, logical, and assignment operators.
- Precedence and associativity of operators.

Input and Output

Input and Output

- Using `System.out.println()` and `System.out.print()`.
- Taking user input using `Scanner` class.



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Introduction to Java

CSC 211



Course Overview

Objective:

The primary goal of this course is to introduce students to the fundamentals of Java programming. By the end of the course, students should be able to write, debug, and understand Java programs, apply object-oriented programming principles, and develop basic Java applications.





Outcomes

- **Proficiency in Java Syntax:** Students will become familiar with Java's syntax and language constructs.
- **Problem-Solving:** Students will develop problem-solving skills through programming exercises.





Outcomes

- **Object-Oriented Programming:** Understanding and applying OOP concepts such as encapsulation, inheritance, and polymorphism.
- **Project Development:** Ability to design and implement small to medium-sized Java projects.





Overview of Java as a Programming Language

- **History and Evolution:** Java was developed by Sun Microsystems in 1995.
- Originally designed for embedded systems, it has since become a widely-used language for building web, mobile, desktop, and enterprise applications.



Features of Java:

- **Object-Oriented:** Java is based on OOP principles, making it modular, reusable, and easy to maintain.
- **Platform-Independent:** Java's "write once, run anywhere" philosophy means that code can run on any device with a Java Virtual Machine (JVM).





Features of Java:

- **Robust and Secure:** Java provides strong memory management, exception handling, and security features.
 - **Multithreaded:** Java supports concurrent processing through multithreading, making it suitable for applications that require parallel execution.
- 



Setting Up Java Development Environment on VS Code

- Visual Studio Code (VS Code) is a lightweight and powerful code editor that supports Java development through various extensions.



Install Java Development Kit (JDK)

- **Step 1:**

- **Download JDK** Visit the official Oracle website or OpenJDK website.

<https://www.oracle.com/java/technologies/downloads/#jdk22-windows>

- Download the latest version of the JDK suitable for your operating system (Windows, macOS, or Linux).

Step 2: Install JDK

- Follow the installation instructions provided on the download page.
- On Windows, during installation, the JDK installer usually sets the `JAVA_HOME` environment variable. If not, you can set it manually.
- On macOS/Linux, you might need to configure the `JAVA_HOME` environment variable in your shell configuration file (e.g., `.bashrc`, `.zshrc`).





Verify the installation by running the command:

`java -version`



Install Visual Studio Code

- **Step 1: Download VS Code**

Visit the [official Visual Studio Code website](https://code.visualstudio.com/)

Download the installer for your operating system (Windows, macOS, or Linux).

- **Step 2: Install VS Code**

Run the installer and follow the setup wizard instructions to install VS Code on your machine.





Install Required Extensions for Java Development

VS Code relies on extensions to support Java development. The most essential ones include:



Java Extension Pack (recommended):

Installation:

1. Open VS Code.
2. Go to the Extensions view by clicking on the Extensions icon in the Activity Bar on the side of the window or by pressing Ctrl+Shift+X.
3. Search for "Java Extension Pack" in the search bar.
4. Click "Install" on the "Java Extension Pack" by Microsoft.

Included Extensions:

- **Language Support for Java(TM) by Red Hat:** Provides basic Java support, including syntax highlighting, IntelliSense (code completion), and more.
- **Debugger for Java:** Allows you to debug your Java applications within VS Code.
- **Java Test Runner:** Facilitates running and debugging Java test cases.
- **Maven for Java:** Integrates Apache Maven with VS Code, making it easier to manage your Java project dependencies.
- **Java Dependency Viewer:** Helps you visualize your project's dependencies and structure.

Additional Useful Extensions:

- **CheckStyle for Java:** Ensures that your code adheres to standard coding practices by providing linting support.
- **Spring Boot Extension Pack:** If you're working on Spring Boot projects, this extension pack provides the necessary tools and features.

Write Your First Java Program

- Open the newly created project folder in VS Code.
- In the src directory, create a new Java file named HelloWorld.java.
- Write the following code

Write Your First Java Program

```
public class Ryekoai {  
    public static void main(String[] args) {  
        System.out.println("My name is Jeff  
Geoff, I love Java Programming!");  
    }  
}
```



Write Your First Java Program

- `public class Ryekoai`: Defines a public class named HelloWorld.

In Java, every application begins with a class definition.

- `public static void main(String[] args)`: This is the entry point of any Java application. The main method is where the program starts execution.

- `System.out.println("My name is Jeff Geoff, I love Java Programming !");`: This line prints the text "Hello, World!" to the console.

Step 3: Run the Java Program

- To run the program, click the play button on the top right corner of the editor or right-click the file and select "Run Java."
- The output should appear in the "Terminal" panel at the bottom of the VS Code window.

Step 4: Debug the Java Program

- Place a breakpoint in your code by clicking in the gutter to the left of the line number.
- Run the debugger by clicking the Run and Debug button or by pressing **F5**.
- Use the Debug panel to step through the code and inspect variables.

